

Error Detection and Correction: Catch the Corrupted Message

Aim

To detect the transmission errors using parity bits in python.

Program

```
import random

# Get message
message = input("Enter a message: ")

# Convert message to binary
binary = ""
for ch in message:
    binary += format(ord(ch), "08b")

print("Binary:", binary)

# Calculate even parity
parity = binary.count("1") % 2
print("Parity Bit:", parity)

# Flip one random bit
binary_list = list(binary)
position = random.randint(0, len(binary_list) - 1)

if binary_list[position] == "0":
    binary_list[position] = "1"
else:
    binary_list[position] = "0"

received = "".join(binary_list)

print("Received Binary:", received)

# Check parity
```

```
new_parity = received.count("1") % 2
```

```
if parity == new_parity:  
    print("OK")  
else:  
    print("ERROR DETECTED")
```

Output:



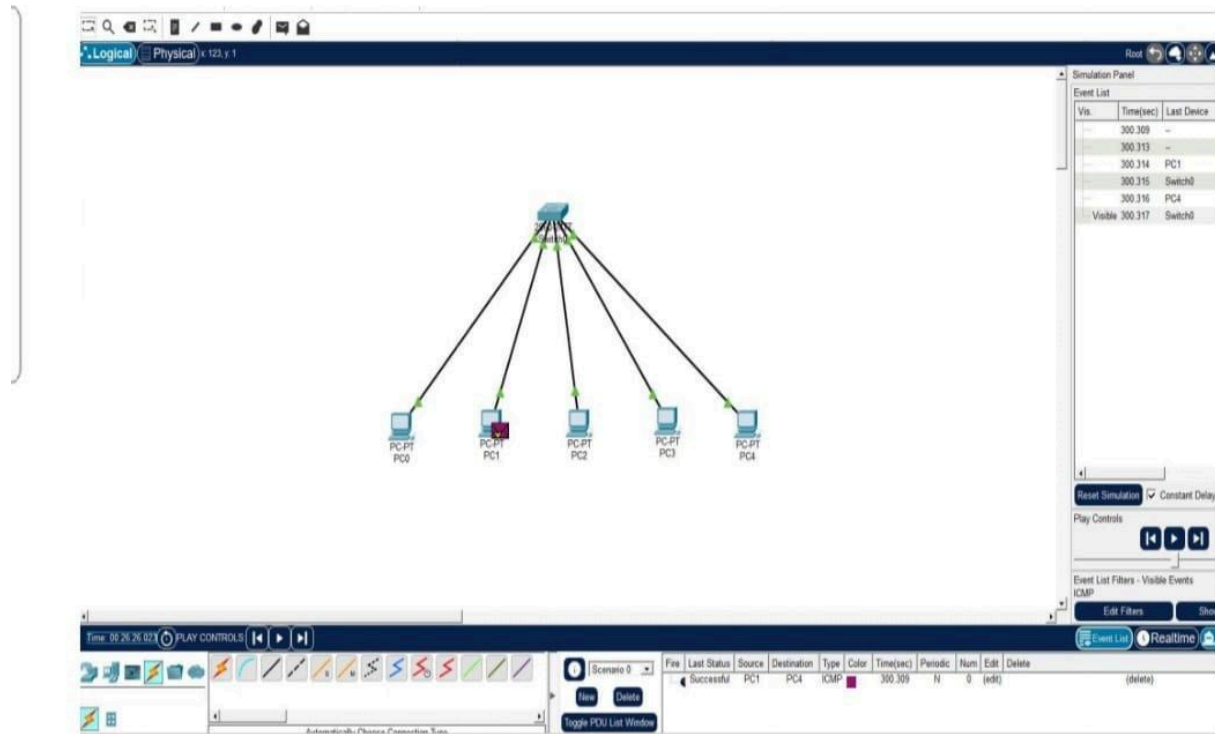
```
IDLE Shell 3.13.5  
File Edit Shell Debug Options Window Help  
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32  
Enter "help" below or click "Help" above for more information.  
= RESTART: C:/Users/91866/AppData/Local/Programs/Python/Python313/Error detection and correction.py  
Enter a message: hii  
Binary: 011010000110100101101001  
Parity Bit: 1  
Received: 011010000010100101101001  
ERROR DETECTED  
Checksum: 314
```

Observation:

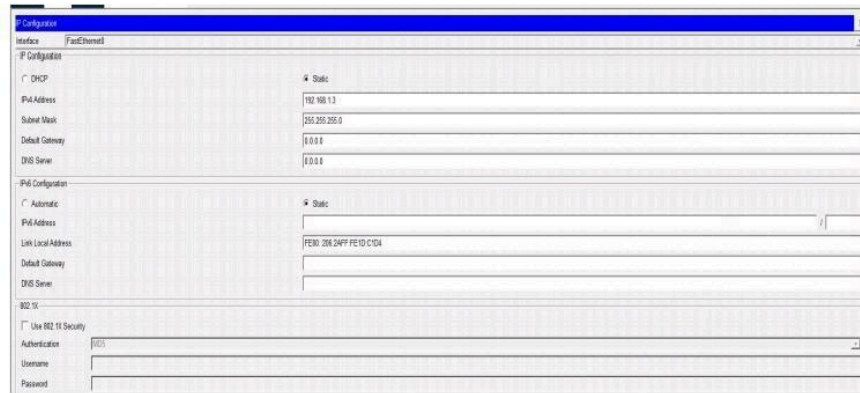
This program converts the input message to 8 bit binary and calculates an even parity bit. It then randomly flips one bit to simulate an error during transmission. This demonstrates single bit Error Detection using parity bit.

LAN AND MULTIPLE ACCESS

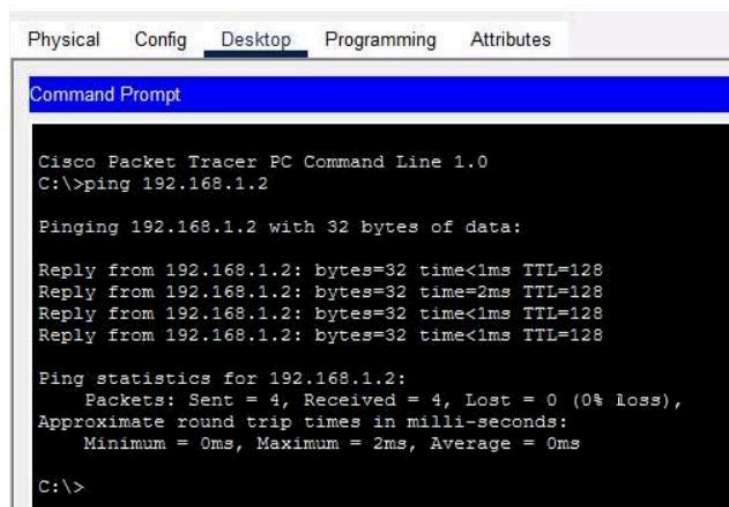
Topology



Configuration



Ping Test



Observation

The network was created using five pcs connected to a switch. IP addresses were assigned to all pc in the same network. Ping testing was performed and communication between the pcs was successful. The switch forwards data only to the destination device which reduces network traffic and collisions. This setup demonstrates efficient LAN communication using a switch.

Wi-Fi Name	Channel	Approx Sign Strength
realme 16 Pro 5G DAE4	6	~ -44 dBm
SIMATS	1, 6, 11	~ -50 to -75 dBm
K13 Turbo Pro 5G	1	~ -52 dBm
lme 15 Pro 5G	1	~ -55 dBm
Shylesh ❤️	6	~ -60 dBm
OnePlus 11 5G	6	~ -62 dBm
motorola	5	~ -65 dBm

Observation

1. Channel 1 is highly congested.
Multiple networks are present with strong signals - 50 to -55 dBm.
2. Channel 6 has strongest signal network.
3. Channel 11 is moderately crowded. All signals are weaker -68 to -75 dBm compared to ch1 and 6.
4. Non overlapping channels 1, 6, 11 are all in use.
Best channel to use is 11.